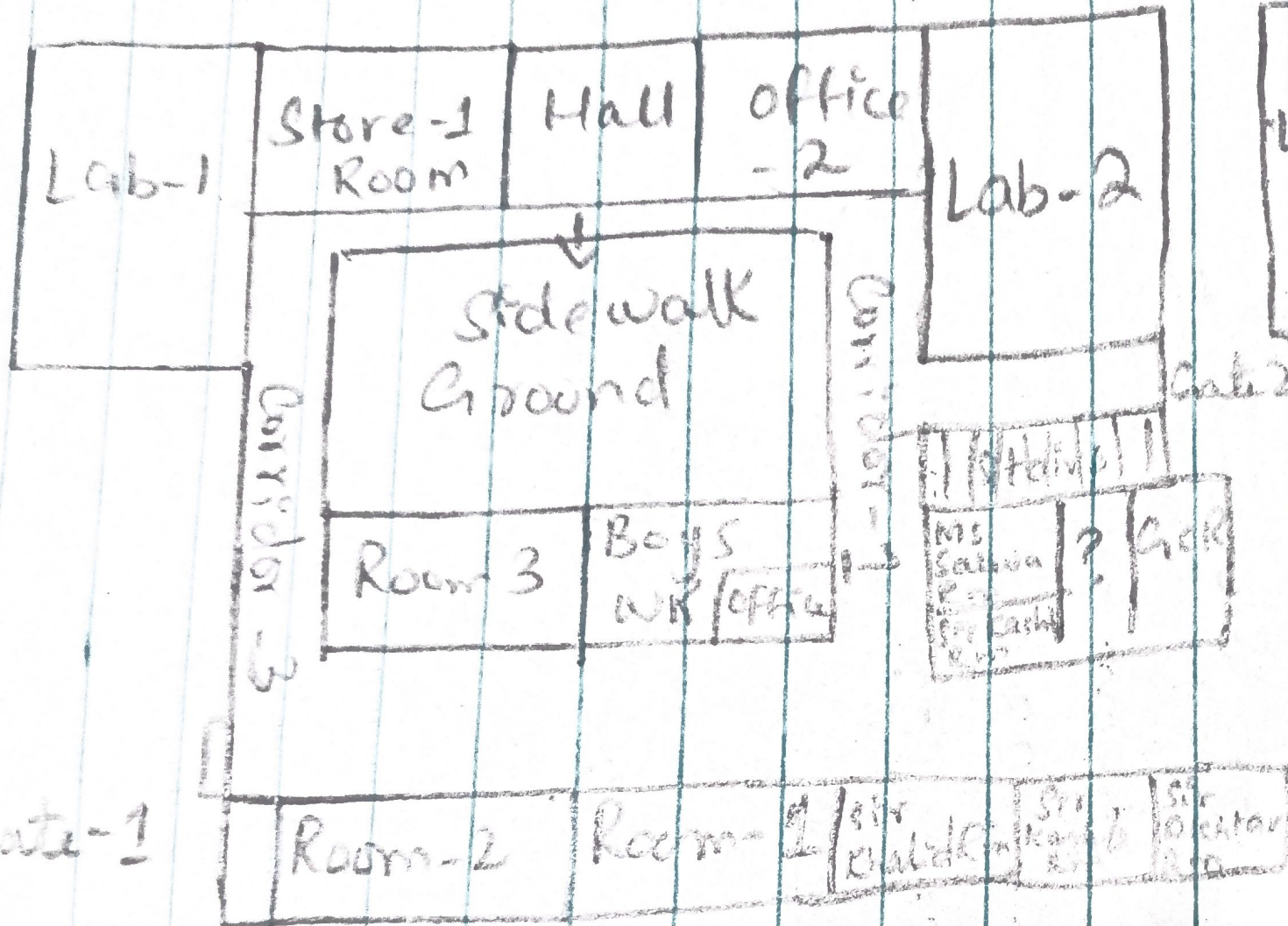
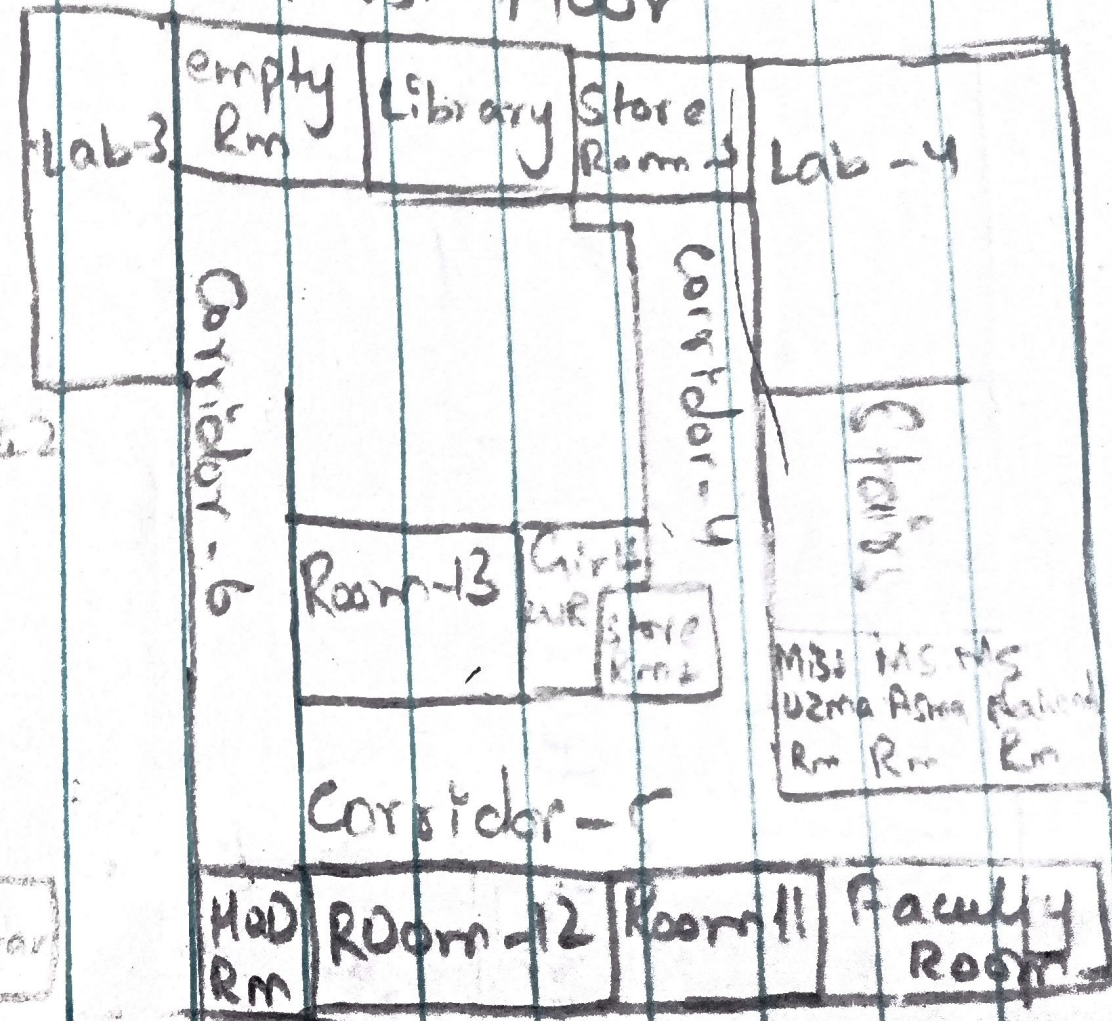


Ground Floor



First Floor



Problem State -

Initial Point:

HOD Room

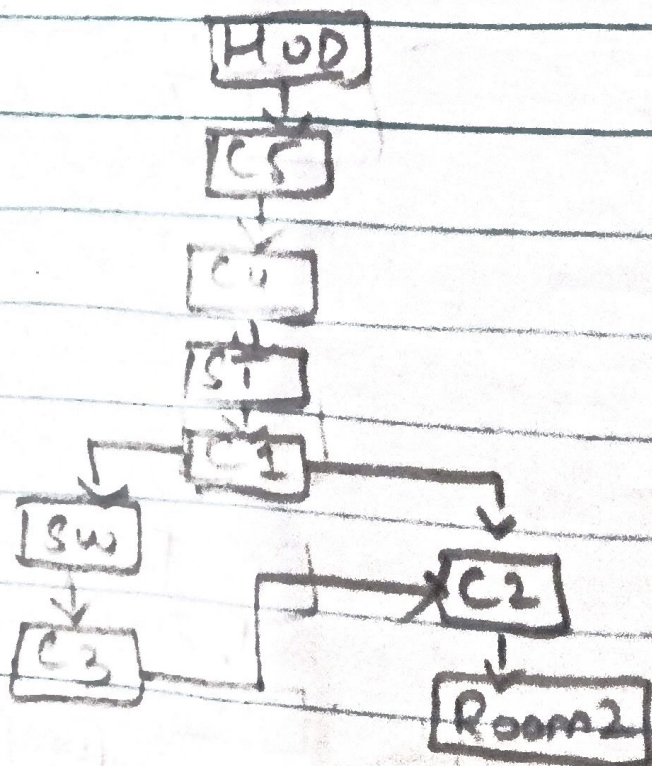
Goal:

Room 2

Graph Representation

Node	connects to
HOD Rm	C5
C5 (Corridor-5)	HOD, C4
C4 (Corridor-4)	C5, ST
ST (Stairs)	C4, C1
C1 (Corridor-1)	ST, C2, SW
C2 (Corridor-2)	C1, Room 2, C3
SW (Side wall)	C1, C3
C3 (Corridor 3)	SW, C2
Room 2	C2

BFS Tree:-



BFS Trace :-

Step 0: Initialize

$Q = [HOD]$

$V = [HOD]$

$S_1 = \text{Deq } HOD \text{ Rm expand } HOD \text{ Rm} \rightarrow (C5)$

$Q = [HOD]$

$V = [HOD]$

$S_2 = \text{Deq } C5 \text{ expand } C5 \rightarrow (HOD, C4)$

HOD room already visited

$Q = [C4]$

$V = [HOD, C5, C4]$

$S_3 = \text{Deq } C4 \text{ expand } C4 \rightarrow (C5, ST)$

$C5$ already visited

$Q = [ST]$

$V = [HOD, C5, C4, ST]$

$S_4 = \text{Deq } ST \text{ expand } ST \rightarrow (C4, C1)$

$C4$ already visited

$Q = [C1]$

$V = [HOD, \dots ST, C1]$

$S_5 = \text{Deq } c_1 \text{ expand } c_1 \rightarrow (c_2, sw, ST)$

ST already visited

$Q = [c_2, sw]$

$V = [HOD, \dots, c_1, c_2, sw]$

$S_6 = \text{Deq } c_2 \text{ expand } c_2 \rightarrow (c_1, Room_2, c_3)$

c_1 already visited

$Q = [sw, Room_2, c_3]$

$V = [HOD, \dots, c_1, c_2, sw, Room_2, c_3]$

$S_7 = \text{Deq } sw \text{ expand } sw \rightarrow (c_1, c_3)$

Both are already visited

$Q = [Room_2, c_3]$

$V = [HOD, \dots, Room_2, c_3]$

$S_8 = \text{Deq } Room_2 \text{ expand } Room_2 (c_2)$

c_2 already visited

$S_9 = \text{Deq } c_3 \text{ expand } c_3 \rightarrow (c_2, sw)$

both are visited

Goal achieved.

Shortest Path:

$HOD \rightarrow c_5 \rightarrow c_4 \rightarrow ST \rightarrow c_1 \rightarrow c_2 \rightarrow Room$ 6 edges / steps

Alternate Path:

$HOD \rightarrow c_5 \rightarrow c_4 \rightarrow ST \rightarrow c_1 \rightarrow sw \rightarrow c_3 \rightarrow c_2 \rightarrow Room_2$
8 edges / steps